TEAM PROJECT • IIT ROPAR

# Predicting Student mental health and depression detection using logistic regression

A machine learning pipeline utilizing logistic regression to analyze student lifestyle factors and predict mental health risks for early depression detection.

## GROUP 60 • TEAM MEMBERS & RESPONSIBILITIES

| MEMBER NAME | Entry Number | ROLE / RESPONSIBILITY |
|---|---|---|
| **Aaditya Sharma** | 2026EEB1310 | Documentation Lead |
| **Divya Goyal** | 2026EEB1348 | Execution Lead |
| **Agrani Soni** | 2026MCB1506 | PPT Lead |
| **Nenavath Yashwanth** | 2026AIB1063 | Video Lead |

**GITHUB REPOSITORY** https://github.com/Divya-iitrpr/Student-Depression-Predictor

01 / CONTEXT

# Core Objective & Dataset

Predicting student depression risks by analyzing key demographic, academic, and lifestyle factors using machine learning.

### PROBLEM STATEMENT

**Student mental health issues often go unnoticed due to subtle lifestyle and academic pressures. This project builds a binary classification model to predict depression risk; categorizing students into Depression or No Depression, enabling early identification and targeted well-being support based on social, health, and academic patterns.**

### WHY IT MATTERS

- Identify key lifestyle patterns and risk factors affecting student mental health.
- Demonstrate the practical application of machine learning in student well-being research.

### DATASET SNAPSHOT

| | |
|---|---|
| **Rows/Columns** | 4464 Records, 13 Columns |
| **Features** | 12 input features + 1 target variable |
| **Source** | Kaggle — https://www.kaggle.com/datasets/mkhoirulmustaqim/student-depression-detection-dataset |

02 / METHOD

# End-to-End Machine Learning Workflow

A structured end-to-end machine learning pipeline from raw student data to depression risk prediction.

## 01 DATA

- Dataset Loading: Load the Student Depression Detection Dataset (4,464 records).
- Exploratory Data Analysis: Understand feature distributions, identify data types, and explore relationships between features and the target.
- Data Preprocessing: Encode categorical variables and scale numerical features appropriately.

## 02 MODEL

- Train-Test Split: Split the data into 80% training and 20% testing using stratification (random_state=19).
- Model Training: Train a Logistic Regression classifier using the preprocessed training data.
- Libraries Used: NumPy, Pandas, Matplotlib, Seaborn, Scikit-Learn.

## 03 OUTPUT

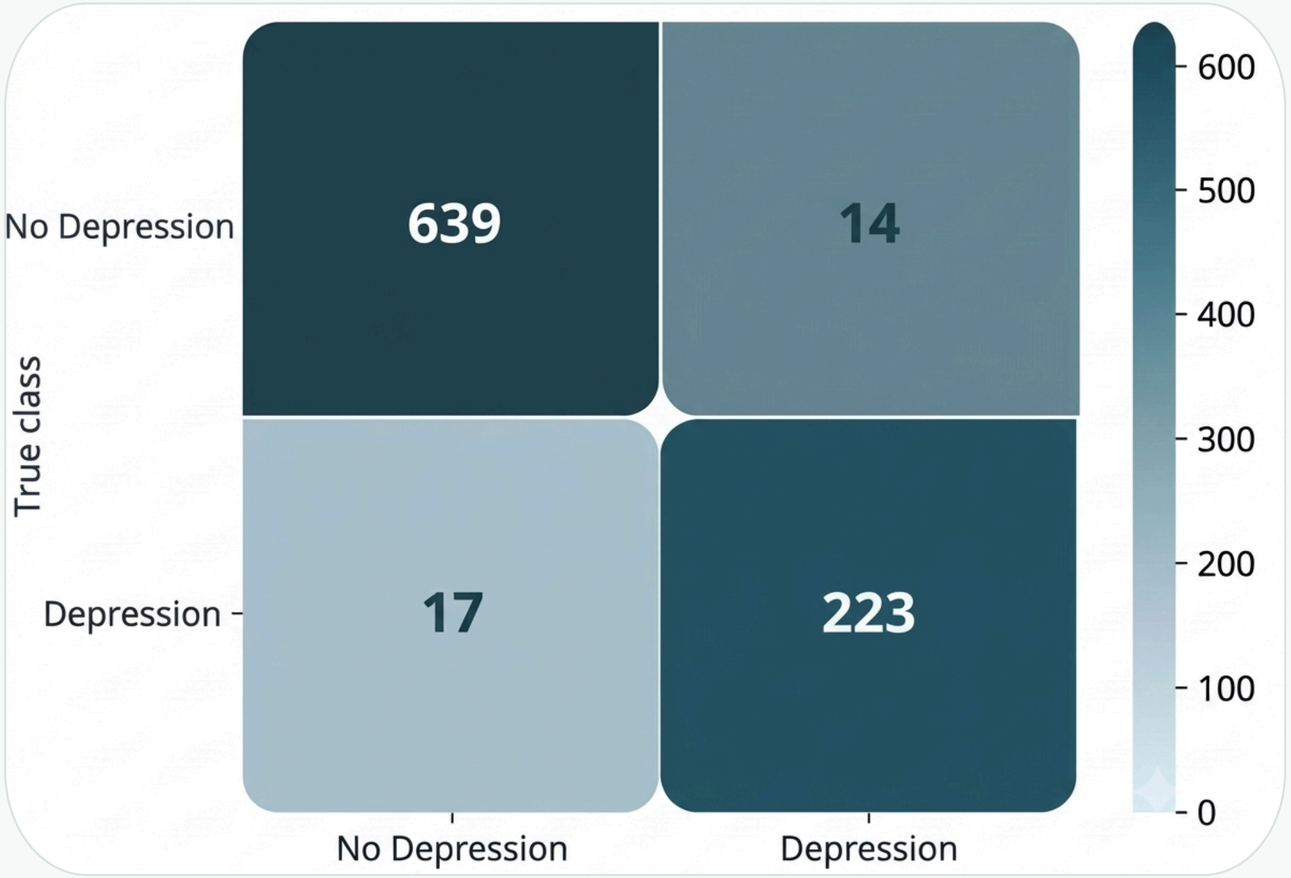
- Prediction: Predict depression categories (Yes / No) on the test data.
- Evaluation & Presentation: Evaluate model performance (metrics like accuracy and confusion matrix) and interpret final results for presentation.

**TOOLS & LIBRARIES** [Python] [Pandas] [NumPy] [Scikit-learn] [Matplotlib]

03 / EVIDENCE

# Model Performance

CONFUSION MATRIX




INTERPRETATION

## Effective Classification

Successfully distinguishes between students with and without depression, achieved 96.53% accuracy.

## Early Identification

Demonstrates the potential of ML in identifying students who may need psychological support.

## Practical Limitation

Serves as a preliminary screening aid, not a substitute for professional diagnosis.

04 / REFLECTION

# Key Technical Takeaways & Workflow Insights

Turning mixed data into a reproducible pipeline while evaluating model limitations responsibly.

### WHAT WORKED

- Mixed-Feature Pipeline: Built a structured preprocessing workflow combining ordinal/one-hot encoding with feature scaling tailored to data types.
- Reproducible Design: Applied stratified splitting and fixed random seeds to ensure full pipeline auditability.

### WHAT DIDN'T

- Single Metric Limitations: relying solely on accuracy was insufficient; evaluating Precision, Recall, F1-Score, and ROC-AUC revealed key trade-offs in risk detection.
- Scope Limitations: Static survey data couldn't fully capture real-time mental health fluctuations or individual student nuances.

### TEAM CONTRIBUTIONS

- Aaditya - Model Training & README
- Divya - Data Exploration & Graphs
- Agrani - Data Preprocessing & PPT
- Yashwanth - Model Testing & Evaluation

### AI-USE DISCLOSURE

• Code Assistance: Used AI for error debugging, syntactical troubleshooting, and exploring model configurations.
• Presentation & Docs: Assisted in structuring slide layouts and refining technical documentation.

**THANK YOU**